Text Editor with Undo and Redo – Stack

Main.py:

main.py

```
from editor import TextEditor


editor = TextEditor()

editor.type_text("Hello ")
editor.type_text("World!")

while True:
    print("\n===== TEXT EDITOR =====")
    print("1. Type Text")
    print("2. Undo")
    print("3. Redo")
    print("4. Display Text")
    print("5. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        new_text = input("Enter text to add: ")
        editor.type_text(new_text)

    elif choice == "2":
        editor.undo()

    elif choice == "3":
        editor.redo()

    elif choice == "4":
        editor.display()

    elif choice == "5":
        print("Text editor closed.")
        break

    else:
        print("Invalid choice.")
```

Editor.py;

editor.py > TextEditor > undo

```python
from stack import Stack

class TextEditor:
    def __init__(self):
        self.text = ""
        self.undo_stack = Stack()
        self.redo_stack = Stack()

    def type_text(self, new_text):
        self.undo_stack.push(self.text)
        self.text = self.text + new_text

        self.redo_stack = Stack()

        print("Text added successfully.")

    def undo(self):
        if self.undo_stack.is_empty():
            print("Nothing to undo.")
            return

        self.redo_stack.push(self.text)
        self.text = self.undo_stack.pop()
        print("Undo successful.")

    def redo(self):
        if self.redo_stack.is_empty():
            print("Nothing to redo.")
            return

        self.undo_stack.push(self.text)
        self.text = self.redo_stack.pop()
        print("Redo successful.")

    def display(self):
        print("\nCurrent text:")
        print(self.text if self.text else "(Empty)")
```

Stack.py:

stack.py > Stack > pop

```python
class Stack:
    def __init__(self):
        self.items = []

    def push(self, item):
        self.items.insert(len(self.items), item)

    def pop(self):
        if self.is_empty():
            return None

        item = self.items[len(self.items) - 1]
        del self.items[len(self.items) - 1]
        return item

    def is_empty(self):
        return len(self.items) == 0
```

Output:

```
PS E:\HEXAWARE TASK\TASK 3\TEXT EDITOR> python main.py
Text added successfully.
Text added successfully.

===== TEXT EDITOR =====
1. Type Text
2. Undo
3. Redo
4. Display Text
5. Exit
Enter your choice: 1
Enter text to add:  Hi
Text added successfully.

===== TEXT EDITOR =====
1. Type Text
2. Undo
3. Redo
4. Display Text
5. Exit
Enter your choice: 4

Current text:
Hello World! Hi

===== TEXT EDITOR =====
1. Type Text
2. Undo
3. Redo
4. Display Text
5. Exit
Enter your choice: 2
Undo successful.
```

```
===== TEXT EDITOR =====
1. Type Text
2. Undo
3. Redo
4. Display Text
5. Exit
Enter your choice: 4

Current text:
Hello World!

===== TEXT EDITOR =====
1. Type Text
2. Undo
3. Redo
4. Display Text
5. Exit
Enter your choice: 3
Redo successful.

===== TEXT EDITOR =====
1. Type Text
2. Undo
3. Redo
4. Display Text
5. Exit
Enter your choice: 4

Current text:
Hello World! Hi

===== TEXT EDITOR =====
1. Type Text
2. Undo
3. Redo
4. Display Text
5. Exit
Enter your choice: 5
Text editor closed.
```